Yichen Xiao

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Education

University of Waterloo Waterloo, ON
Bachelors of Applied Science, Computer Engineering *Sep. 2026 – May 2031*

Experience

Full-Stack Web Developer Oct. 2025 – Present
St. Theresa of Lisieux CHS *Richmond Hill, ON*

- Architected and built a full-stack athletics data management platform using Next.js and Supabase, replacing a legacy website prone to errors and serving over 1600 student athletes across 32 sports teams
- Designed a relational Postgres schema tracking team rosters, individual athlete performance, and payment collection to generate reports directly informing end-of-year banquet funding decisions and award nominations
- Developed a batch email communication system using Nodemailer enabling admins to select recipients and send instructions, updates, and notifications to over 40 coaches, eliminating repetitive one-by-one email outreach
- Optimized data table performance using Tanstack Table, implementing pagination and client-side filtering to reduce load times by 70% when browsing large data views

Captain, Lead Builder and Programmer June 2024 – Apr. 2026
VEX Robotics Competition Teams 82855S and 82855X *Richmond Hill, ON*

- Captained a 9-member robotics team, leading CAD in Fusion 360, mechanical assembly, and programming for 82855S; previously coded for 82855X, earning Provincial Skills Champion and a World Championship qualification
- Designed and tuned PID systems for motion and arm mechanism control using PROS and LemLib, replacing manual motor velocity regulation and improving autonomous routine consistency by 50%
- Built a particle filter localization algorithm in C++, fusing IMU, distance sensor, and deadwheel odometry data to address positional drift beyond standard odometry

Software Education Lead and Chief Financial Officer July 2025 – Apr. 2026
Lions Robotics *Richmond Hill, ON*

- Designed and delivered a structured curriculum covering C++ fundamentals, software-hardware integration, and autonomous localization and path-planning algorithms
- Provided one-on-one mentorship to new members, assisting with development environment setup and debugging
- Managed a $25,000+ budget, coordinating with sponsors to secure funding for hardware, tools, and competitions

Projects

Checkmate | *C++*

- Built a 2500+ ELO chess engine in C++ from scratch, implementing minimax search algorithms with alpha-beta pruning, custom evaluation functions, and perfect-hash functions for move generation
- Optimized search performance through search heuristics and use of stack-allocated move lists, achieving 20M+ nodes per second in move generation
- Utilized cache-friendly data structures with contiguous memory layout, leveraging 64-bit bitwise operations for board state manipulation that compile to single CPU instructions, improving search efficiency by 90%

King of the Ring | *HTML/CSS, Javascript — 1st place at Eureka Hacks 2026*

- Built a real-time, motion-controlled boxing game using browser-based using browser-based gyroscope and accelerometer APIs for gesture recognition
- Implemented a Node.js server using Websockets, managing room-code-based matchmaking and relaying real-time motion data between mobile phones and the game display, achieving sub-100 millisecond latency
- Contributed to rule-based gesture classification, processing accelerometer and gyroscope signal data and helping tune detection thresholds to distinguish between 9 distinct boxing motions

Technical Skills

Languages: C++, JavaScript, Typescript, HTML/CSS, Python, Java, SQL (Postgres), MongoDB
Frameworks: React, Next.js, Node.js, TailwindCSS
Developer Tools: Git, Google Cloud Platform, VS Code, Claude Code
Libraries: NumPy, PyTorch, Matplotlib, Pandas, PROS, LemLib, Java Swing